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Monocytes in type 1 diabetes families exhibit high cytolytic activity and subset abundances that correlate with clinical progression

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Monocytes are immune regulators implicated in the pathogenesis of type 1 diabetes (T1D), an autoimmune disease that targets insulin-producing pancreatic β cells. We determined that monocytes of recent onset (RO) T1D patients and their healthy siblings express proinflammatory/cytolytic transcriptomes and hypersecrete cytokines in response to lipopolysaccharide exposure compared to unrelated healthy controls (uHCs). Flow cytometry measured elevated circulating abundances of intermediate monocytes and >2-fold more $CD14^+CD16^+HLADR^+KLRD1^+PRF1^+$ NK-like monocytes among patients with ROT1D compared to uHC. The intermediate to nonclassical monocyte ratio among ROT1D patients correlated with the decline in functional β cell mass during the first 24 months after onset. Among sibling nonprogressors, temporal decreases were measured in the intermediate to nonclassical monocyte ratio and NK-like monocyte abundances; these changes coincided with increases in activated regulatory T cells. In contrast, these monocyte populations exhibited stability among T1D progressors. This study associates heightened monocyte proinflammatory/cytolytic activity with T1D susceptibility and progression and offers insight to the age-dependent decline in T1D susceptibility.

INTRODUCTION

Type 1 diabetes (T1D) is a heterogeneous autoimmune disease that results in the loss of pancreatic β cell functional mass and lifelong dependence on exogenous insulin replacement therapy. As evidenced by the partial successes of immunotherapies targeting T cells (*1*, *2*), T1D pathogenesis involves adaptive immune responses. However, a growing body of evidence supports a role for altered innate immune responses and inflammation in T1D pathogenesis (*3*–*6*).

Monocytes are innate immune cells that comprise ~10% of the nucleated blood cell count. They are often classified into three phenotypically and functionally distinct subgroups based on the relative expression of CD14 [lipopolysaccharide (LPS) receptor] and CD16 [Fc gamma receptor III (FcγRIII)] (*7*). Classical monocytes ($CD14^+CD16^-$), which comprise 80 to 85% of the peripheral pool in heathy individuals, are efficient phagocytes and can secrete inflammation-stimulating factors (*8*). A portion of classical monocytes differentiate into the intermediate subtype ($CD14^+CD16^+$), which comprise 2 to 8% of the circulating pool. They are characterized by higher major histocompatibility complex (MHC) class II expression and function in antigen presentation, T cell stimulation, and the production of reactive oxygen species. Intermediate monocytes can further differentiate into the nonclassical subset ($CD14^{dim}CD16^+$), which comprise 2 to 11% of circulating monocytes. While nonclassical monocytes can function in a proinflammatory capacity and liberate interleukin-1 (IL-1), tumor necrosis factor (TNF), and IL-12 in response to infection, they also function in the resolution of inflammation (*9*, *10*), maintenance of vascular homeostasis, and the clearance of dying cells, viruses, and tumor cells from the circulation (*11*).

Monocytes have been implicated in the pathology of numerous inflammatory and autoimmune diseases (*12*). Expansions of intermediate monocytes have been reported in Graves' disease (*13*), rheumatoid arthritis (*14*), and anti-neutrophil cytoplasmic antibody–associated vasculitis (*15*), while nonclassical monocytes have been implicated in relapsing-remitting multiple sclerosis, coronary artery disease, and systemic lupus erythematosus (*11*). Compared to unrelated healthy controls (uHCs), Ren *et al.* (*16*) reported that patients with recent onset T1D (ROT1D) had a greater abundance of circulating intermediate monocytes that was positively associated with hemoglobin A1c (HbA1c) and negatively associated with serum insulin and C-peptide levels. Further, Bradshaw *et al.* (*17*) reported that monocytes of patients with T1D spontaneously secrete IL-1β and IL-6, inflammatory cytokines known to induce and expand T helper 17 (T_H17) cells, suggesting that abnormal innate immune activity may drive adaptive responses in T1D by expanding the T_H17 population of effector T cells.

The advancement of single-cell RNA sequencing (scRNA-seq) has provided insight into the cellular heterogeneity of the immune system. Using this approach, Villani *et al.* (*18*) described four monocyte subsets in healthy human participants, where the intermediate monocyte subset was separated into two groups, designated mono3 and mono4 (*18*). Notably, the mono4 subset expressed a cytolytic transcriptome, which included high expression of perforin, granzymes, and granulysin. Using scRNA-seq, a similar monocyte subset has also been identified by Hu *et al.* (*19*).

On the basis of the work summarized above, as well as our own studies (*3*, *20*, *21*), we hypothesized that altered monocyte abundances and activity may be associated with T1D progression. We therefore examined monocytes isolated from patients with ROT1D, healthy autoantibody-negative siblings of probands, and uHC. Monocytes of T1D family members were found to hypersecrete cytokines upon

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stimulation, and among patients with ROT1D, most monocyte subsets expressed a proinflammatory/cytolytic transcriptional signature that shared identity with the mono4 subpopulation (*18*). Flow cytometry revealed elevated mono4 and intermediate monocyte abundances among patients with ROT1D, and the intermediate-to-nonclassical monocyte ratio was found to negatively correlate with residual β cell functional mass during the post-onset period. In contrast to individuals that progressed to stage 1 T1D (≥2 autoantibodies/normoglycemia), stage 2 T1D (≥2 autoantibodies/dysglycemia), or stage 3 T1D (clinical onset), the abundance of mono4 monocytes longitudinally decreased among nonprogressors and their abundances negatively correlated with temporal increases in peripheral activated T_{reg} cells. These analyses identify heightened monocyte inflammatory activity associated with T1D susceptibility and progression that is temporally counter-regulated among nonprogressors, offering insight to the age-dependent decline in T1D susceptibility.

RESULTS

Monocytes of T1D family members hypersecrete cytokines in response to TLR4 ligation

Hussain *et al.* (*22*) reported that peripheral blood mononuclear cell (PBMC) isolated from patients with T1D and their healthy first-degree relatives hypersecrete IL-1α and TNF-α after mitogen stimulation. To further characterize innate immune function in T1D families, we isolated monocytes from cryopreserved PBMCs drawn from ROT1D, autoantibody-negative siblings of probands with high-risk (DR3 and/or DR4) human leukocyte antigen (HLA) genotypes (HRS) or with lower risk (non-DR3/DR4) HLA genotypes (LRS), and uHC. The characteristics of the study participants are provided in table S1. Monocytes were cultured in the presence of LPS, and the secretion of IL-1β, IL-6, and TNF-α was measured in the conditioned medium after 24 hours. Levels of cytokine secretion were not different among the patients with ROT1D and the autoantibody-negative healthy sibling cohorts. By contrast, LPS-stimulated IL-1β, IL-6, and TNF-α secretion was elevated in ROT1D monocytes, as well as the normoglycemic HRS and LRS cohorts when compared to uHC ($P < 0.05$; Fig. 1, A to C).

Transcriptional analysis of negatively selected monocytes using Affymetrix arrays

The transcriptomes of monocytes negatively selected from cryopreserved PBMCs of uHC, LRS, HRS, and patients with ROT1D were analyzed. We identified 826 probe sets that met a threshold of $|\log_2 \text{ratio}| > 0.5$ and an false discovery rate (FDR) < 20% among the six possible pairwise comparisons (table S2). Overall, volcano plots showed symmetry in each of the pairwise comparisons (fig. S1). Hierarchical clustering of this dataset determined that the monocytes of the autoantibody-negative, normoglycemic HRS and LRS groups exhibited intermediate relatedness to those of uHC and ROT1D, which were highly distinct from one another (Fig. 2A).

Elevated abundances of intermediate monocytes have previously been reported among patients with T1D (*16*). We hypothesized that the 826 probe sets differentially regulated among the four cohorts would reflect increasing differentiation towards the intermediate and nonclassical subsets. Therefore, the 826 probe set signature of ROT1D, HRS, LRS, and uHC monocytes were compared to the dataset of Schmidl *et al.* (*23*), which examined the transcriptomes of sorted $CD14^{++}CD16^{-}$ (classical), $CD14^{+}CD16^{+}$ (intermediate),

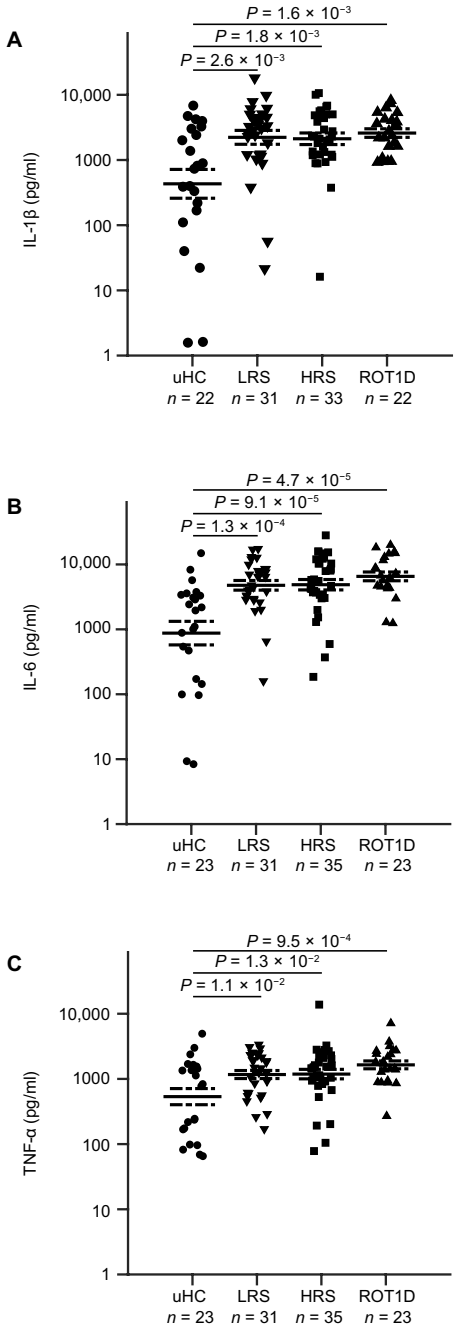


Fig. 1. Monocytes of patients with ROT1D and their healthy autoantibody negative siblings are hyper-responsive to TLR4 ligation compared to uHC. The analysis included uHC [(A) $n = 22$; age, 15.9 ± 2.7 years; (B and C) $n = 23$; age 15.7 ± 2.9 years], stage 0 LRS [(A) $n = 31$; age, 10.9 ± 3.3 years; (B and C) $n = 31$; age, 10.8 ± 3.4 years; all autoantibody negative], stage 0 HRS [(A) $n = 33$; age, 11.4 ± 3.6 years; (B and C) $n = 35$; age, 11.3 ± 3.6 years; all autoantibody negative], and patients with ROT1D [(A) $n = 22$; age, 11.2 ± 3.7 years; HbA1c, 7.3 ± 1.0%; (B and C) $n = 23$; age, 11.0 ± 3.8 years; HbA1c, 7.3 ± 1.0%]. Monocytes were negatively selected from PBMCs. A total of 10^6 monocytes per well were incubated in RPMI complete medium for 24 hours with LPS (10 ng/ml). The conditioned medium was recovered, and cytokines (IL-1β, IL-6, and TNF-α) were measured in duplicate by ELISA [(**A** to **C**), respectively]. Each dot represents a single participant. Horizontal bars indicate the mean ± SE. Significance was assessed with a two-tailed *t* test. Cytokine secretion levels between patients with ROT1D and the healthy normoglycemic sibling cohorts were not different ($P > 0.05$).

and $CD14^{dim}CD16^{+}$ (nonclassical) monocytes. Among the transcripts that exhibited differential expression among the monocyte subgroups in the Schmidl dataset, we identified 2359 unique transcripts (based on the Human Genome Organization gene symbol) that were also present on the Affymetrix U133 plus 2 array. Among this subset, a relatively small but significantly overlapping set of 126 unique transcripts (χ^2, $P < 1.0 \times 10^{-5}$, 177 probe sets) were differentially expressed among ROT1D, HRS, LRS, and uHC monocytes. As reflected by the heatmaps (Fig. 2B) and Pearson's correlation coefficients (Fig. 2C), for these commonly regulated transcripts, uHC monocytes were most correlated with the classical subset and exhibited high expression of transcripts previously associated with classical monocytes (Fig. 2E). The HRS monocytes were most correlated with the intermediate subset, and the ROT1D monocytes were most correlated with the nonclassical subset.

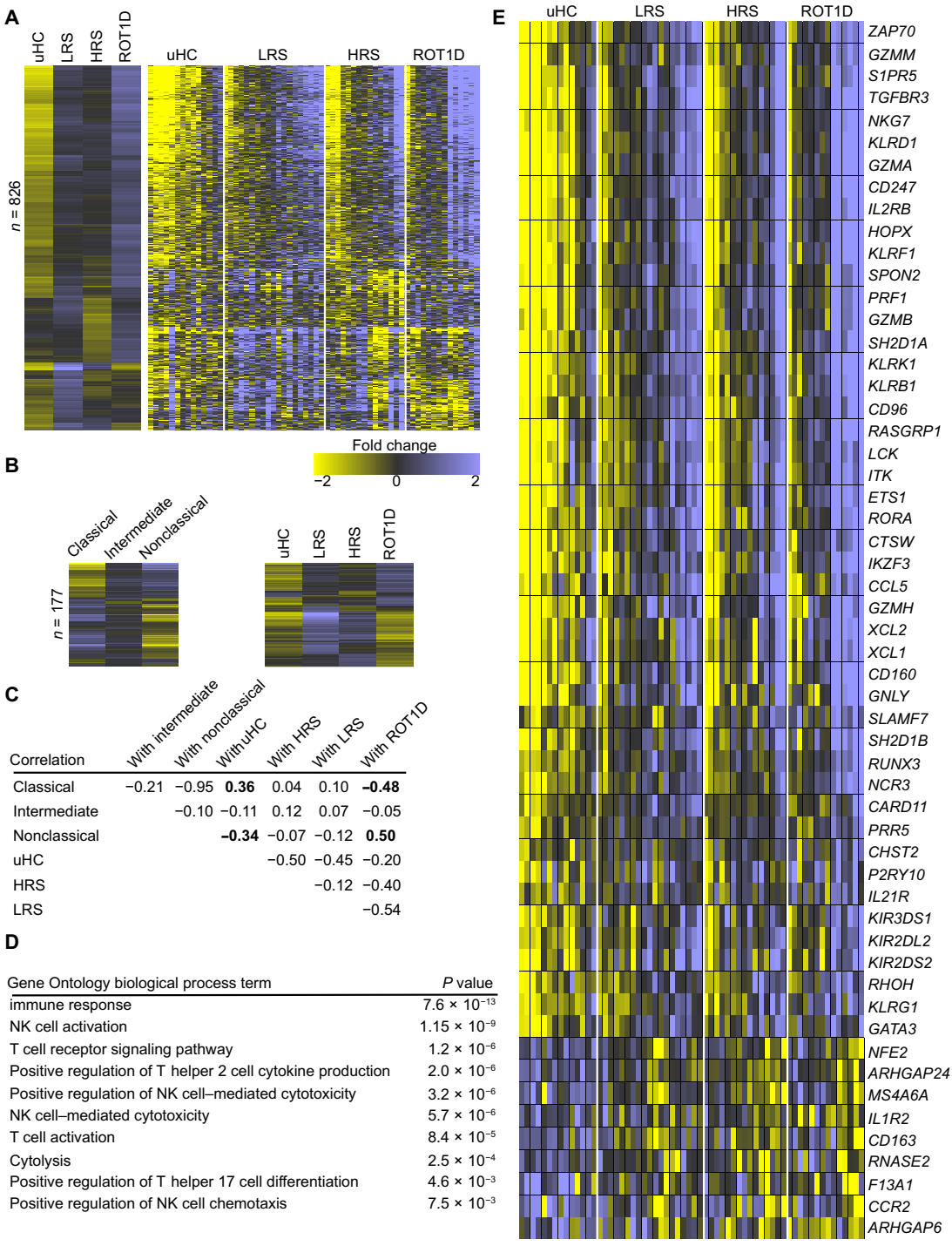
To further characterize the 826 probe sets differently regulated among ROT1D, HRS, LRS, and uHC monocytes, Gene Ontology (GO) analyses were conducted (Fig. 2D). Consistent with prior reports that have associated increased abundances of intermediate monocytes with T1D (*16*), the ROT1D monocytes expressed a transcriptional signature consistent with antigen presentation/T cell costimulation (*CD160*, *GZMA*, *ITK*, *LCK*, *NFATC2*, *NLRC3*, *RORA*, *SLAMF6*, *SLAMF7*, and *ZAP70*). The ROT1D monocyte signature was also enriched in transcripts related to natural killer (NK) cell activation (*GATA3*, *IL2RB*, *IL18R1*, *IL21R*, *KIR2DS2*, *KIR3DS1*, *NCR1*, *NCR3*, and *SH2D1A*) and NK cytotoxicity (*CD7*, *CD247*, *CTSW*, *FASLG*, *GNLY*, *GZMB*, *IL18RAP*, *KIR3DL2*, *KLRB1*, *KLRD1*, *NKG7*, *PRF1*, and *SH2D1B*). A similar, albeit less robust signature was observed in the healthy sibling cohorts. The cytolytic ROT1D signature overlapped with that described for the mono4 intermediate subset defined by Villani *et al.* (*18*), which also exhibited high expression of *PRF1*, *GZMB*, *GNLY*, *CTSW*, and *KLRD1*. Specifically, Villani *et al.* (*18*) identified 149 transcripts associated with mono4; 132 of these were represented on the Affymetrix U133 plus 2 array, and 69 of 132 (52.3%; χ^2, $P < 1.0 \times 10^{-5}$) were among the 826 transcripts differentially expressed among monocytes of the uHC, LRS, HRS, and ROT1D cohorts.

scRNA-seq analysis of ROT1D, HRS, LRS, and uHC monocytes

Because scRNA-seq has fostered the identification of monocyte populations extending beyond the traditional tripartite classification scheme (*18*, *19*), we applied it to monocyte-enriched PBMC populations of uHC, LRS, HRS, and ROT1D study participants. For these eight participants (two per cohort), 40,000 cells were sequenced at a depth of 500 to 5000 transcripts per cell. The 2000 genes exhibiting the highest expression variability among these cells were subjected to principal components analysis (PCA). For visualization, the top 50 significant PCs were reduced to two dimensions using *t*-distributed stochastic neighbor embedding (t-SNE); this unsupervised analysis identified 13 clusters (Fig. 3A). Differential expression analysis was performed for each cluster (table S3), and the identity of each cluster was defined by comparing significantly associated marker gene expression (adjusted $P < 0.001$) to that of established reference clusters in blood samples. The following putative populations were identified: $CD14^{+}$ monocytes, NK cells, naïve $CD4^{+}$ T cells, memory $CD4^{+}$ T cells, naïve $CD8^{+}$ T cells, NKT cells, B cells, $CD16^{+}$ monocytes, memory $CD8^{+}$ T cells, γδ T cells, $CD4^{+}$ regulatory T (T_{reg}) cells, CD1c dendritic cells, and plasmacytoid dendritic cells (Fig. 3, A to C).

The $CD14^{high}$ monocyte cluster exhibited high expression of numerous transcripts previously associated with the classical and intermediate subsets (*CD14*, *CSTA*, *CTSS*, *CXCL8*, *MNDA*, *MS4A6A*, *S100A8*, *S100A9*, *S100A12*, *VCAN*, *LGALS1*, *LGALS2*, and *LYZ*) (*19*). The $CD16^{high}$ monocyte cluster exhibited high expression of transcripts previously associated with the nonclassical subset (*FCGR3A*, *FCGR3B*, *TCF7L2*, *MS4A7*, *RHOC*, *IFITM3*, *CD79B*, *CSF1R*, *CDKN1C*, *HES4*, *SMIM25*, *MTSS1*, *BCL2A1*, and *RRAS*) (*19*). We examined the transcriptional profiles across the clusters and detected gene expression consistent with antigen presentation (*HLA-DPA1*, *HLA-DPB1*, *HLA-DQB1*, *HLA-DRB1*, and *HLA-DMA*) by monocytes as well as B cells and dendritic cells. Expected gene expression patterns were also observed for NK cells [*KLRD1*, *KLRF1*, *NKG7*, and *NCAM1*(*CD56*)], naïve $CD4^{+}$ T cells (*CCR7*, *FHIT*, and *CD4*), memory $CD4^{+}$ T cells (*TNFRSF4*, *IL7R*, and *CD40LG*), naïve $CD8^{+}$ T cells (*CD8b* and *LEF1*), NKT cells (*NCR3* and *KLRB1/CD161*), B cells (*CD79A* and *CD79B*), memory $CD8^{+}$ T cells (*CD8A*, *CCL5*, and *IL32*), γδ T cells (*TRGC2* and *NUCB2*), $CD4^{+}$ T_{reg} cells (*FOXP3*, *CTLA4*, and *IL2RA*), CD1c dendritic cells (*CD1C*, *CLEC10A*, and *PDL4*), and plasmacytoid dendritic cells (*CLSPN*, *RAD51AP1*, and *STMN1*) (*18*, *19*, *24*).

The $CD14^{high}$ and $CD16^{high}$ monocyte clusters illustrated in Fig. 3 totaled 13,915 cells; these were subjected to further analysis. The top 10 significant PCs were used to reduce the data to two dimensions and visualized by Uniform Manifold Approximation and Projection (UMAP). High heterogeneity was observed as evidenced by the identification of 10 clusters (Fig. 4A), all of which expressed *CD14* and/or *CD16* as well as *HLA-DRB1* (Fig. 4, A to C). Azimuth analysis predicted that 99.11% (13,791 of 13,915) of these cells were monocytes (*25*). The clusters were characterized by their top marker genes (table S4) and through ontological analysis to identify their associated GO biological processes and molecular functions (table S5). Cluster 3 exhibited the highest *CD14* expression, low *CD16* expression, and consistent with studies of sorted classical monocytes (*23*) and expressed the highest levels of *EIF2AK2*, *IFI44*, *IFI44L*, *MX1*, and other transcripts related to antiviral and innate immune responses. Cluster 5 exhibited the highest *CD16* and lowest *CD14* abundance transcript levels, as well as the highest levels of previously reported marker genes for nonclassical monocytes (*FCGR3A*, *CDKN1C*, *HES4*, *TCF7L2*, *RHOC*, and *MTSS1*) (*23*, *26*). In line with this observation, in silico pseudotime trajectory analysis of the 10 clusters (fig. S2) found cluster 5 the most differentiated. The remaining clusters expressed varying levels of *CD14* and *CD16*, suggesting differing transition states from classical to nonclassical monocytes. Paralleling the S100A12 subset described by Hu *et al.* (*19*), cluster 0 expressed high levels of the proinflammatory calgranulin genes *S100A8*, *S100A9*, and *S100A12*. In line with the high *S100* and *HMGB2* transcript expression, RAGE receptor binding was a top molecular function associated with cluster 0. Ribosomal proteins are essential for ribosome synthesis, which in turn are required for cell growth, differentiation, and proliferation. While other clusters also expressed numerous transcripts encoding ribosomal proteins and transcripts related to cytoplasmic translation (*EEF1A1*, *RPS2*, *RPL18A*, *RPS3A*, *RPL15*, and *RPS18*), this was the most distinguishing feature of cluster 1 and was consistent with the identification of cytoplasmic translation being identified as the top GO Term. The top marker genes for cluster 2 included *CLEC4E*, *VCAN*, *VIM*, *JUN*, and *FPR1*, which were co-annotated under the GO term protein binding. Clusters 4 and 8 expressed the highest



C

Correlation	With intermediate	With nonclassical	With uHC	With HRS	With LRS	With ROT1D
Classical	−0.21	−0.95	**0.36**	0.04	0.10	**−0.48**
Intermediate		−0.10	−0.11	0.12	0.07	−0.05
Nonclassical			**−0.34**	−0.07	−0.12	**0.50**
uHC				−0.50	−0.45	−0.20
HRS					−0.12	−0.40
LRS						−0.54

D

Gene Ontology biological process term	*P* value
immune response	7.6×10^{-13}
NK cell activation	1.15×10^{-9}
T cell receptor signaling pathway	1.2×10^{-6}
Positive regulation of T helper 2 cell cytokine production	2.0×10^{-6}
Positive regulation of NK cell–mediated cytotoxicity	3.2×10^{-6}
NK cell–mediated cytotoxicity	5.7×10^{-6}
T cell activation	8.4×10^{-5}
Cytolysis	2.5×10^{-4}
Positive regulation of T helper 17 cell differentiation	4.6×10^{-3}
Positive regulation of NK cell chemotaxis	7.5×10^{-3}

Fig. 2. Monocytes of patients with ROT1D and their healthy autoantibody negative siblings exhibit proinflammatory transcriptomes compared to uHCs. The analysis included uHC ($n = 14$; age, 15.3 ± 2.8 years), stage 0 LRS ($n = 19$, age, 11.7 ± 2.8 years, all autoantibody negative), stage 0 HRS ($n = 15$; age, 12.9 ± 3.4 years, all autoantibody negative), and patients with ROT1D ($n = 14$; age,12.4 ± 2.8 years; HbA1c, 7.5 ± 1.4%). RNA extracted from monocytes negatively selected from cryopreserved PBMCs was analyzed with the Affymetrix U133plus 2.0 array. This array interrogates 54,613 transcripts/variants. Because low intensity data are often unreliable, only probe sets where ≥75% of the samples had a $\log_2$ intensity > 4.5 relative fluorescence units (RFU) ($n = 31{,}571$) were retained for analysis. Among the six pairwise comparisons, 826 probe sets met thresholds: $|\log_2 \text{ratio}| > 0.5$, FDR < 20%. The number of transcripts meeting thresholds for each pairwise comparison: ROT1D versus HRS = 215, ROT1D versus LRS = 9, ROT1D versus uHC = 309, LRS versus uHC = 206, LRS versus HRS = 122, and HRS versus uHC = 356. (**A**) Hierarchical clustering of the 826 probe set union. (**B**) Heatmap illustrating the mean expression levels of 177 commonly regulated unique probe sets ($|\log_2 \text{ratio}| > 0.5$; FDR < 0.2; χ^2 $P < 1.0 \times 10^{-5}$) among sorted monocyte subsets described by Schmidl *et al.* (*23*), and the 826 differentially expressed transcripts by ROT1D, HRS, LRS, and uHC monocytes. (**C**) Pearson correlation coefficients for the 177 commonly regulated unique probe sets. (**D**) GO biological processes associated with the 826 probe set union. (**E**) Expression levels of well-annotated transcripts that met thresholds among the six-pairwise comparisons.

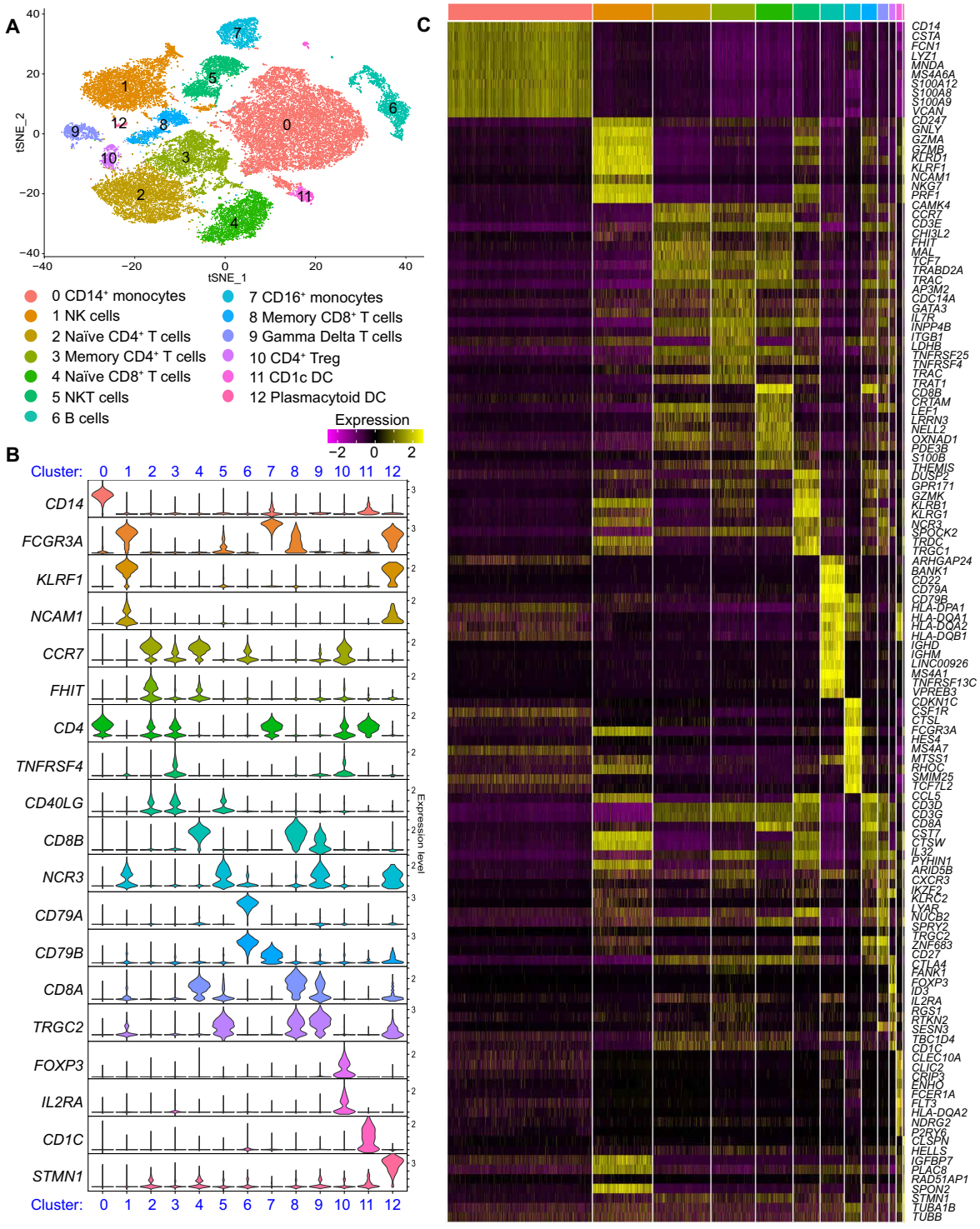
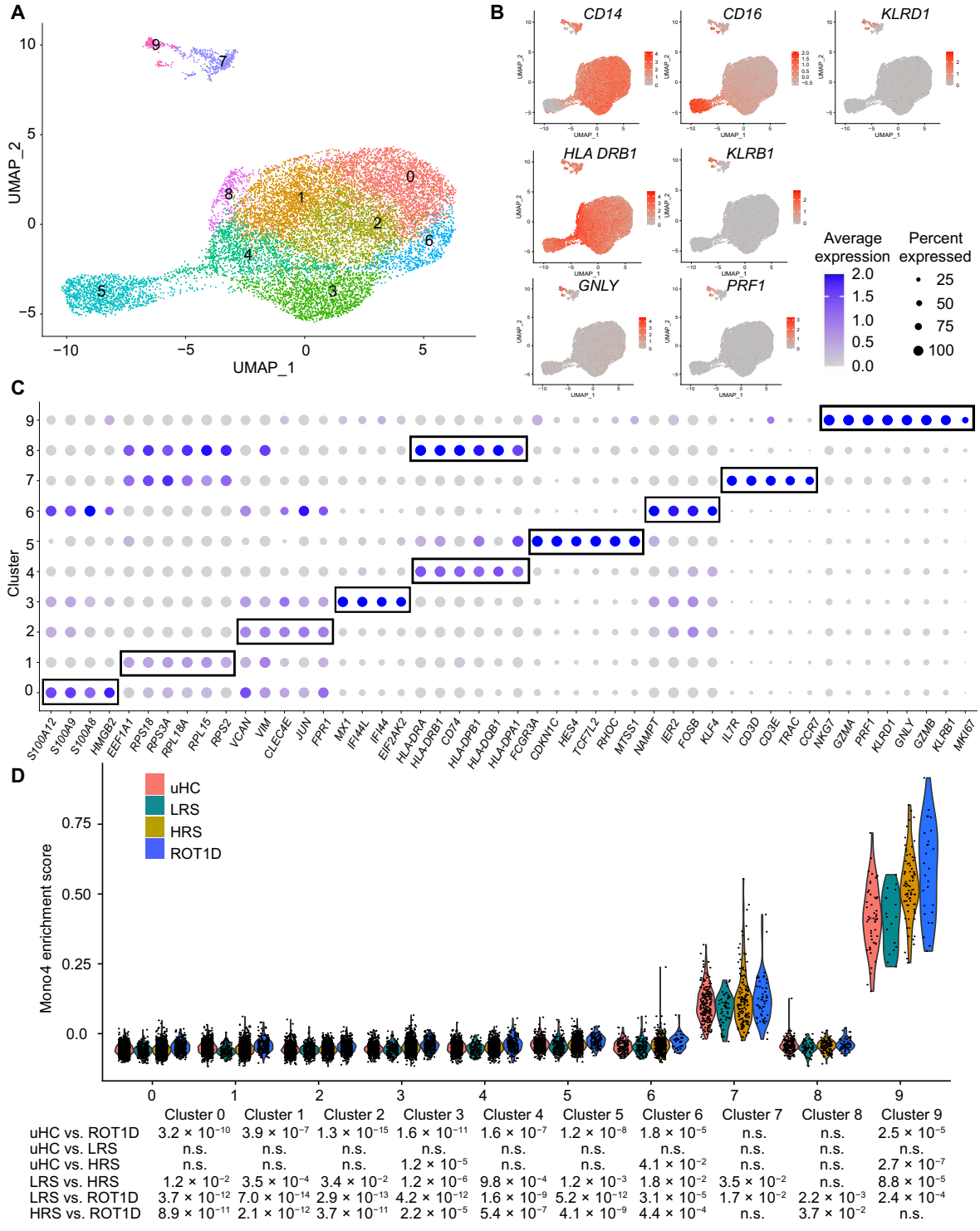


Fig. 3. scRNA-seq analysis of monocyte-enriched PBMCs of uHC, LRS, HRS, and patients with ROT1D. The analysis included uHC ($n = 2$; age, 15.9 ± 2.6 years), stage 0 LRS ($n = 2$; age, 10.7 ± 0.2 years, all autoantibody negative), stage 0 HRS ($n = 2$; age, 13.6 ± 3.8 years, all autoantibody negative), and patients with ROT1D ($n = 2$; age, 9.4 ± 1.5 years; HbA1c, 7.6 ± 0.8%). (**A**) t-SNE plot shows 38,594 sequenced cells distributed into 13 clusters. Colors denote cluster identity as determined by the unsupervised clustering algorithm, and each dot represents an individual cell. The putative cluster identities, based on expression of established cell lineage markers, are indicted. The number of successfully profiled single cells per cluster were: $CD14^+$ monocytes ($n = 12{,}542$), NK cells ($n = 5154$), naïve $CD4^+$ T cells ($n = 4973$), memory $CD4^+$ T cells ($n = 3782$), naïve $CD8^+$ T cells ($n = 3168$), NKT cells ($n = 2262$), B cells ($n = 2013$), $CD16^+$ monocytes ($n = 1373$), memory $CD8^+$ T cells ($n = 1324$), γδ T cells ($n = 894$), $CD4^+$ T_{reg} cells ($n = 522$), CD1c dendritic cells ($n = 481$), and plasmacytoid dendritic cells ($n = 106$). (**B**) Violin plots illustrate expression distribution of lineage-defining marker genes across the 13 clusters. (**C**) A heatmap illustrating the scaled expression of up to 15 discriminative gene sets (adjusted P value $< 3.5 \times 10^{-12}$) for each of the 13 clusters. The color bar illustrates the z-score distribution from −2.5 (purple) to 2.5 (yellow). DC, dendritic cell.



	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7	Cluster 8	Cluster 9
uHC vs. ROT1D	3.2×10^{-10}	3.9×10^{-7}	1.3×10^{-15}	1.6×10^{-11}	1.6×10^{-7}	1.2×10^{-8}	1.8×10^{-5}	n.s.	n.s.	2.5×10^{-5}
uHC vs. LRS	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
uHC vs. HRS	n.s.	n.s.	n.s.	1.2×10^{-5}	n.s.	n.s.	4.1×10^{-2}	n.s.	n.s.	2.7×10^{-7}
LRS vs. HRS	1.2×10^{-2}	3.5×10^{-4}	3.4×10^{-2}	1.2×10^{-6}	9.8×10^{-4}	1.2×10^{-3}	1.8×10^{-2}	3.5×10^{-2}	n.s.	8.8×10^{-5}
LRS vs. ROT1D	3.7×10^{-12}	7.0×10^{-14}	2.9×10^{-13}	4.2×10^{-12}	1.6×10^{-9}	5.2×10^{-12}	3.1×10^{-5}	1.7×10^{-2}	2.2×10^{-3}	2.4×10^{-4}
HRS vs. ROT1D	8.9×10^{-11}	2.1×10^{-12}	3.7×10^{-11}	2.2×10^{-5}	5.4×10^{-7}	4.1×10^{-9}	4.4×10^{-4}	n.s.	3.7×10^{-2}	n.s.

Fig. 4. scRNA-seq profiling of circulating monocytes. (**A**) UMAP plot shows 13,915 monocytes were clustered into ten subsets. Colors denote cluster identity as determined by unsupervised clustering algorithm, and each dot represents an individual cell. Numbers of successfully profiled single cells per subset are as follows: cluster 0 ($n = 2798$), cluster 1 ($n = 2451$), cluster 2 ($n = 2302$), cluster 3 ($n = 1872$), cluster 4 ($n = 1480$), cluster 5 ($n = 1300$), cluster 6 ($n = 702$), cluster 7 ($n = 440$), cluster 8 ($n = 397$), and cluster 9 ($n = 173$). (**B**) Expression pattern of marker genes (*CD14*, *CD16/FCGR3A*, *GNLY*, *PRF1*, *KLRD1*, and *KLRB1*) are indicated by color intensity. (**C**) Dot plot shows relative expression of selected marker genes belonging to each cluster. (**D**) Gene module scores for clusters 0 to 9, calculated using the AddModuleScore function in Seurat, were used to quantify the average expression level of the 97 mono4/NK-like transcripts at the single-cell level for each cohort. Tabulated are the *P* values for module score differences between cohorts that reached significance. n.s., not significant.

and most diverse levels of HLA-related transcripts (*HLA-DRA*, *HLA-DRB1*, *CD74*, *HLA-DPB1*, *HLA-DQB1*, and *HLA-DPA1*), and the top GO terms associated with these clusters included "Antigen processing and presentation of exogenous peptide antigen via MHC class II" consistent with a robust antigen processing/presenting capacity for these intermediate subsets. Cluster 6 was characterized by transcripts related to positive regulation of transcription from RNA polymerase II promoter (*KLF4*, *FOSB*, *IER2*, and *NAMPT*). Both Villani *et al.* (*18*) and Hu *et al.* (*19*) previously identified a monocyte subset that exhibited highly distinct clustering from the other monocyte subsets and expressed a cytolytic gene expression profile. Similarly, we identified corresponding cells distributed primarily within two clusters. Consistent with studies showing T cell receptor expression by neutrophils, monocytes and macrophages (*27*, *28*), cluster 7 expressed numerous transcripts related to the T cell receptor complex and immune activation (*CD3D*, *TRAC*, *CD3E*, *IL7R*, and *CCR7*). Further, 21 of the top 100 upregulated cluster 7 marker genes were shared with the 149 transcripts expressed by the mono4 subset defined by Villani *et al.* (*18*). Cluster 9 expressed a cytolytic transcriptional gene signature (*GZMA*, *GZMB*, *GNLY*, and *PRF1*) that shared numerous expressed transcripts with NK cells (*KLRB1*, *KLRD1*, and *NKG7*), indicating overlapping functions between this population and NK cells. Further, 50 of the top 100 upregulated cluster 9 marker genes were shared with the 149 transcripts expressed by the mono4 subset defined by Villani *et al.* (*18*). In silico pseudotime analysis indicated a distinct differentiation trajectory for clusters 7 and 9 relative to the other clusters (fig. S2).

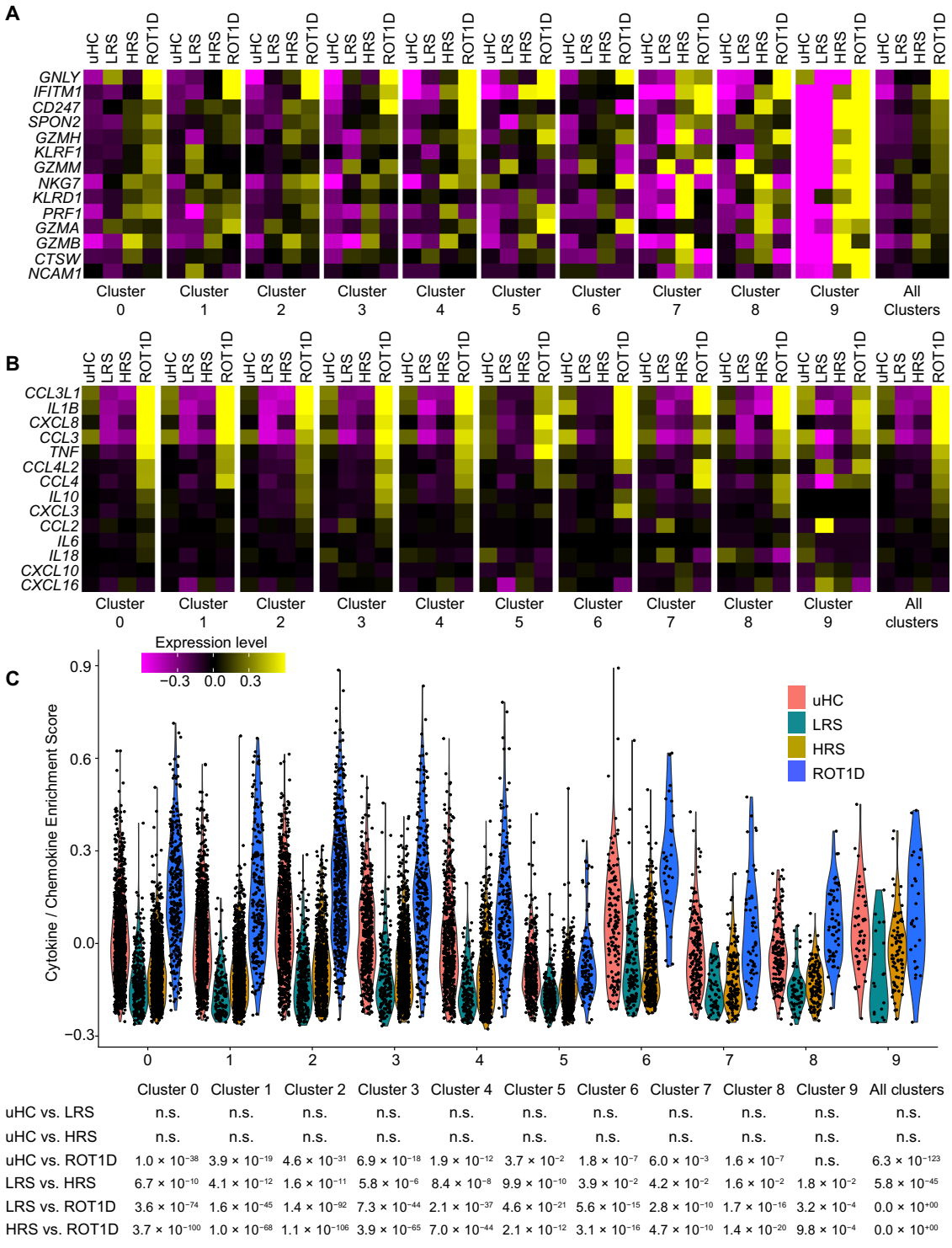
To quantify the average expression levels of the mono4/NK-like transcripts at the single-cell level, we calculated gene module scores for monocyte clusters 0 to 9 (Fig. 4D). This used the intersection of transcripts differentially expressed in the array analysis between patients with ROT1D and uHC and those significantly expressed by the mono4 cluster defined by Villani *et al.* (*18*) and/or the NK-like cluster defined by Hu *et al.* (*19*). For each cluster, this set of 97 overlapping transcripts was used to determine a module enrichment score for monocytes of the uHC, LRS, HRS, and ROT1D cohorts. Consistent with their shared transcript content with mono4, clusters 7 and 9 exhibited higher enrichment scores, indicating higher expression of these proinflammatory and cytolytic transcripts. Notably, compared to uHC and LRS monocytes, a significantly higher enrichment score was observed among patients with ROT1D for cluster 9. Compared to uHC, LRS, and HRS monocytes, a significantly higher enrichment score was also observed among patients with ROT1D for clusters 0 to 6. This observation suggests that monocytes of patients with ROT1D in general express a cytolytic transcriptome that shares identity with that of NK cells.

To further investigate the expression of cytolytic transcripts among the uHC, LRS, HRS, and ROT1D cohorts, differential expression analysis was conducted for the top 100 transcripts in cluster 9. As shown in Fig. 5A, *GNLY*, *PRF1*, *NKG7*, and other NK-like transcripts are generally expressed at the highest levels among patients with ROT1D, at intermediate levels among the sibling cohorts, and at the lowest levels among uHC monocytes. We also examined differential transcript expression of chemokines and cytokines known to be produced by monocytes; 14 mediators were identified that exhibited significant differences when comparing total monocytes of the four cross-sectional cohorts (Fig. 5, B and C). Cytokine/chemokine expression was highest among patients with ROT1D. Consistent with our prior reports describing a state of regulated susceptibility among healthy siblings of patients with T1D (*29*), circulating monocytes of the LRS and HRS cohorts expressed lower levels of these mediators.

Flow cytometric analysis of monocyte populations

To confirm and extend the transcriptomic data, we began by examining the abundances of classical, intermediate, and nonclassical monocytes in the cross-sectional cohorts by flow cytometry after surface staining for CD14 and CD16 (Fig. 6A). Because hyperglycemia is known to affect many immune and inflammatory markers, these analyses included long-standing (LS) T1D patient controls (age, 12.7 ± 5.4 years after onset) that exhibited glycemic control similar to the ROT1D cohort (HbA1c, 8.5 ± 1.0% versus 7.8 ± 1.3%, respectively; $P > 0.05$, two-tailed t test). We measured a higher abundance of classical $CD14^+CD16^-$ monocytes among total monocytes when comparing patients with ROT1D to uHC (85.3% versus 76.4%, $P = 0.01$) and patients with ROT1D to LST1D patients (85.3% versus 74.1%, $P = 0.0001$); classical $CD14^+CD16^-$ monocytes were also more abundant when comparing HRS to uHC (84.7% versus 76.4%, $P = 0.04$; Fig. 6B). Consistent with Ren *et al.* (*16*), we measured a higher abundance of intermediate $CD14^+CD16^+$ monocytes among total monocytes when comparing patients with ROT1D to uHC (10.5% versus 6.4%, $P = 0.01$). The abundance of intermediate $CD14^+CD16^+$ monocytes among total monocytes in the LST1D cohort was similar to the ROT1D group (12.4% versus 10.5%, $P > 0.05$); however, this population was more abundant among patients with LST1D compared to HRS, LRS, and uHC (Fig. 6B). The abundances of classical and intermediate monocytes in the LRS and HRS groups were between the levels measured for the ROT1D and uHC groups. The percentages of nonclassical $CD14^{dim}CD16^+$ monocytes among total monocytes were lower among patients with ROT1D compared to uHC (4.1% versus 17.2%, $P = 0.0001$); they were also lower when comparing the healthy siblings to uHC (LRS versus uHC: 9.1% versus 17.2%, $P = 0.025$; HRS versus uHC: 6.8% versus 17.2%, $P = 0.001$) (Fig. 6B). The patients with LST1D exhibited percentages of nonclassical $CD14^{dim}CD16^+$ monocytes among total monocytes that were similar to the uHC patients and notably higher than HRS and patients with ROT1D (LST1D versus HRS: 13.3% versus 6.8%, $P = 0.043$; LST1D versus ROT1D: 13.3% versus 4.1%, $P = 0.0003$). NK cells were defined as being $CD3^-CD19^-CD56^+CD14^-$, and their abundances among total lymphocytes across the five cohorts did not significantly differ (Fig. 6C).

To identify and enumerate mono4 monocytes, we developed an antibody panel to target surface and intracellular proteins that were among the top cluster 9 marker genes (Fig. 4C), as well as among the highly expressed transcripts of (i) the negatively selected ROT1D monocytes (Fig. 2E), (ii) the mono4 population described by Villani *et al.* (*18*), and (iii) the NK-like monocyte population described by Hu *et al.* (*19*). We selected as cell-surface targets the lectin-like receptors KLRD1 (CD94) and KLRB1 (CD161). KLRD1 expression by NK cells is associated with effector function and cell survival and is linked to innate and adaptive immune responses to pathogens (*30*). Notably, KLRB1 expression identifies a proinflammatory NK cell population that exhibits high cytokine responsiveness and may contribute to inflammatory disease pathogenesis (*31*). We selected the cytolytic protein perforin (PRF1) as an intracellular target. Monocytes of uHC, LRS, HRS, and ROT1D study participants were analyzed by flow cytometry using an antibody panel that targeted CD14, CD16, CD3, CD19, CD56, HLA-DR, KLRB1, KLRD1, and PRF1 (Fig. 7A).



	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7	Cluster 8	Cluster 9	All clusters
uHC vs. LRS	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
uHC vs. HRS	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
uHC vs. ROT1D	1.0×10^{-38}	3.9×10^{-19}	4.6×10^{-31}	6.9×10^{-18}	1.9×10^{-12}	3.7×10^{-2}	1.8×10^{-7}	6.0×10^{-3}	1.6×10^{-7}	n.s.	6.3×10^{-123}
LRS vs. HRS	6.7×10^{-10}	4.1×10^{-12}	1.6×10^{-11}	5.8×10^{-6}	8.4×10^{-8}	9.9×10^{-10}	3.9×10^{-2}	4.2×10^{-2}	1.6×10^{-2}	1.8×10^{-2}	5.8×10^{-45}
LRS vs. ROT1D	3.6×10^{-74}	1.6×10^{-45}	1.4×10^{-92}	7.3×10^{-44}	2.1×10^{-37}	4.6×10^{-21}	5.6×10^{-15}	2.8×10^{-10}	1.7×10^{-16}	3.2×10^{-4}	$0.0 \times 10^{+00}$
HRS vs. ROT1D	3.7×10^{-100}	1.0×10^{-68}	1.1×10^{-106}	3.9×10^{-65}	7.0×10^{-44}	2.1×10^{-12}	3.1×10^{-16}	4.7×10^{-10}	1.4×10^{-20}	9.8×10^{-4}	$0.0 \times 10^{+00}$

Fig. 5. All scRNA-seq clusters of ROT1D monocytes express elevated levels of cytolyic and cytokine/chemokine transcripts. (**A**) Expression levels of cytolytic transcripts differentially expressed between monocytes of uHC, LRS, HRS, and patients with ROT1D. Differential expression analysis was conducted for the top 100 transcripts in cluster 9. Shown are transcripts that meet a threshold of $P < 0.05$ in any one of the six possible pairwise comparisons. (**B**) Expression levels of selected cytokines and chemokines differentially expressed between monocytes of uHC, LRS, HRS, and patients with ROT1D. The cytokines and chemokines shown met an adjusted *P* value of <0.05 in one or more of the six possible comparisons between cohorts. Shown are monocytes of all clusters merged and each individual monocyte cluster (0 to 9). (**C**) Chemokine/cytokine gene enrichment scores for clusters 0 to 9, calculated using the AddModuleScore function in Seurat, were used to quantify the average expression level of the 14 cytokine/chemokine transcripts at the single-cell level for each cohort.

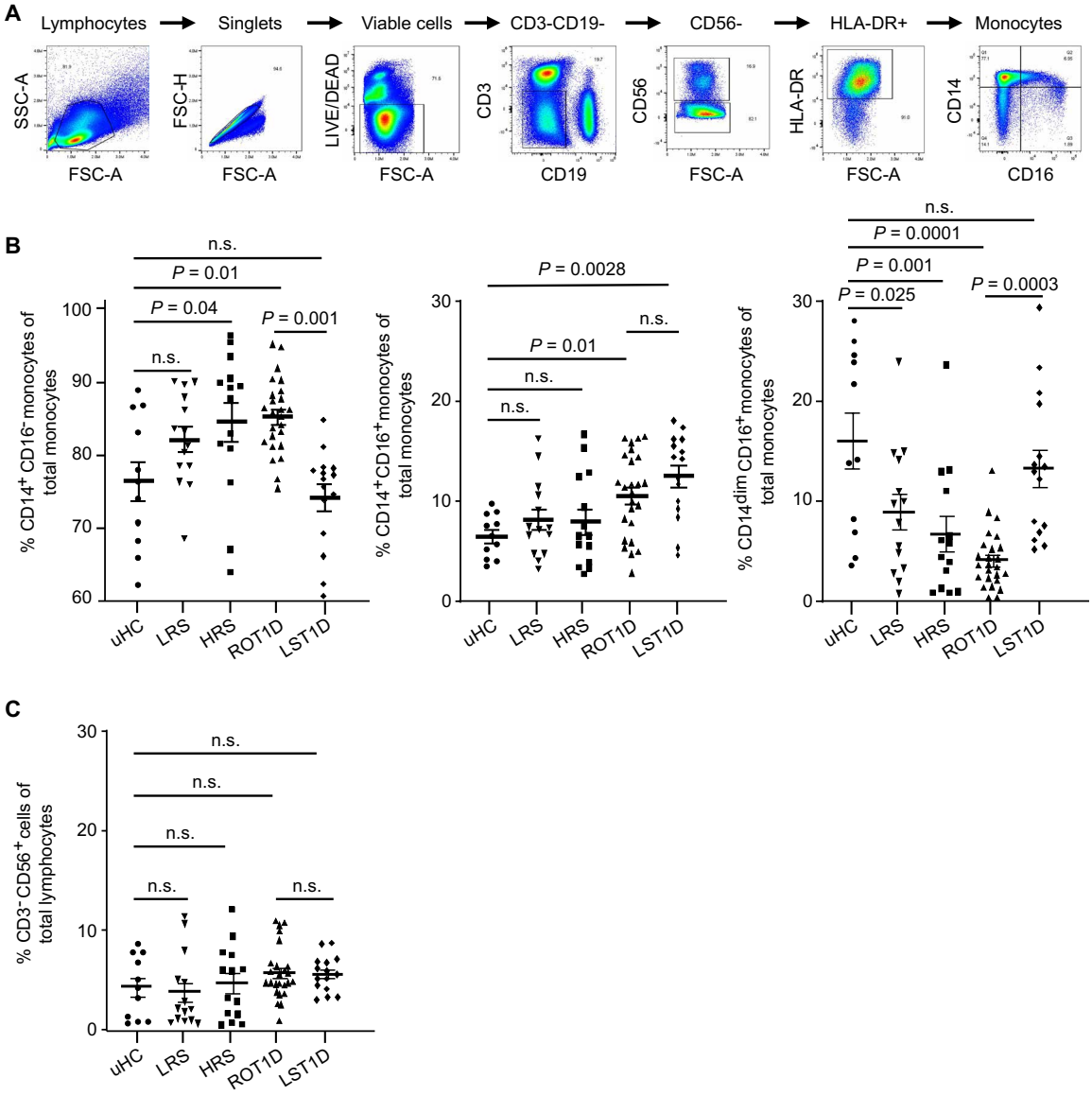


Fig. 6. Cross-sectional flow cytometry analyses show patients with ROT1D have higher circulating percentages of intermediate monocytes. The analysis included uHC ($n = 11$; age, 14.0 ± 2.8 years), stage 0 LRS ($n = 14$; age, 12.1 ± 3.8 years; all autoantibody negative), stage 0 HRS ($n = 14$; age, 12.4 ± 3.2 years; all autoantibody negative), patients with ROT1D ($n = 25$; age, 12.3 ± 3.0 years; HbA1c, 7.8 ± 1.3%), and patients with LST1D ($n = 15$; age, 18.7 ± 8.0 years; HbA1c, 8.5 ± 1.0%). (**A**) Representative flow cytometry profiles for identification of classical, intermediate, and nonclassical monocytes. Cryopreserved PBMCs were thawed and stained. A negative gating strategy was used to identify $CD3^-$, $CD19^-$, and $CD56^-$ lymphocytes. Monocytes were differentiated from NK cells by gating on $HLA\text{-}DR^{hi}$ and segregated into $CD14^+CD16^-$ (classical), $CD14^+CD16^+$ (intermediate), and $CD14^{dim}CD16^+$ (nonclassical) populations. (**B**) Percentages of classical, intermediate, and nonclassical monocytes in the uHC, LRS, HRS, ROT1D, and LST1D cohorts (left to right). (**C**) Percentages of NK cells in the uHC, LRS, HRS, ROT1D, and LST1D cohorts. The differences between groups were assessed by analysis of variance (ANOVA).

Compared to uHC, the classical $CD14^+CD16^-$ and intermediate $CD14^+CD16^+$ monocytes of LRS, HRS, and ROT1D participants had an increasing abundance of $KLRD1^{high}KLRB1^{high}$ monocytes; this difference reached statistical significance when comparing the uHC and ROT1D cohorts. When considering the total monocyte population (classical + intermediate + nonclassical), the overall abundance of $KLRD1^{high}KLRB1^{high}$ monocytes among patients with ROT1D was significantly higher than uHC (3.1% versus 1.4%, $P = 0.04$; Fig. 7B). Similar relationships were identified when analyzing $KLRD1^{high}PRF1^{high}$ monocytes (1.5% versus 0.3%, $P = 0.03$; Fig. 7D) and $KLRD1^{high}GZMB^{high}$ monocytes (3.3% versus 0.5%, $P = 0.04$). The abundances of the mono4 populations among patients with LST1D (Fig. 7, B to E)—as defined by $KLRD1^{high}KLRB1^{high}$, $KLRD1^{high}PRF1^{high}$, or $KLRD1^{high}GZMB^{high}$—were similar to the abundances observed in uHC and the healthy sibling cohorts, supporting the notion that the elevated abundances of this population in the ROT1D cohort was not related to glycemic control.

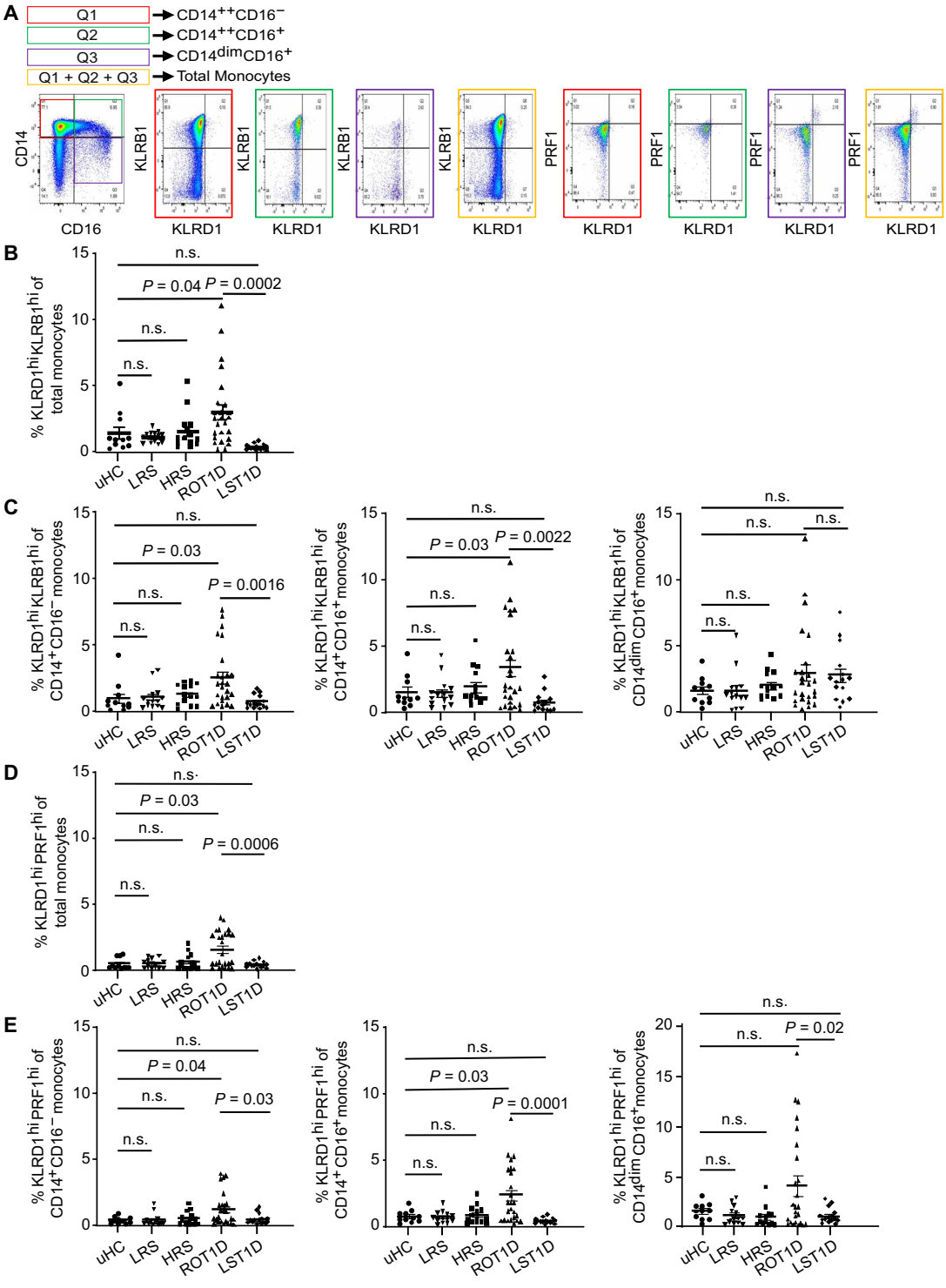


Fig. 7. Cross-sectional flow cytometry analyses show patients with ROT1D have higher circulating percentages of mono4 monocytes. The analysis included uHC ($n = 11$; age, 14.0 ± 2.8 years), stage 0 LRS ($n = 14$; age, 12.1 ± 3.8 years, all autoantibody negative), stage 0 HRS ($n = 14$; age, 12.4 ± 3.2 years; all autoantibody negative), patients with ROT1D ($n = 25$; age, 12.3 ± 3.0 years; HbA1c, 7.8 ± 1.3%), and patients with LST1D ($n = 15$; age, 18.7 ± 8.0 years; HbA1c, 8.5 ± 1.0%). (**A**) Representative flow cytometry profiles for identification of KRLD1^{high}KLRB1^{high} and KLRD1^{high}PRF1^{high} monocytes. (**B**) Percentages of KLRD1^{high}KLRB1^{high} monocytes among total monocytes among the uHC, LRS, HRS, ROT1D, and LST1D cohorts. (**C**) Percentages of KLRD1^{high}KLRB1^{high} monocytes among classical, intermediate, and nonclassical monocytes (left to right) among the uHC, LRS, HRS, ROT1D, and LST1D cohorts. (**D**) Percentages of KLRD1^{high}PRF1^{high} monocytes among total monocytes among the uHC, LRS, HRS, ROT1D, and LST1D cohorts. (**E**) Percentages of KLRD1^{high}PRF1^{high} monocytes among classical, intermediate, and nonclassical monocytes (left to right) among the uHC, LRS, HRS, ROT1D, and LST1D cohorts. Each dot represents a different individual. Horizontal bars indicate the mean ± SE. The differences between groups were assessed by ANOVA.

We found that the transcriptomes of uHC monocytes positively correlated with the gene expression profiles of sorted classical monocytes, while transcriptomes of ROT1D monocytes positively correlated with the gene expression profiles of sorted nonclassical monocytes (Fig. 2C). Consistent with this observation, the lowest percentage of KLRD1, KLRB1, and PRF1 expressing classical and intermediate monocytes was observed in the uHC cohort. Overall, these data were consistent with transcriptomic analyses in that they showed significantly higher mono4 abundances in patients with ROT1D and detected the mono4/NK-like population distributed across the classical, intermediate, and nonclassical subsets.

Relationships between monocyte populations and β cell function among patients with ROT1D

Mixed meal tolerance tests (MMTTs) permit calculation of stimulated C-peptide AUC (area under the curve), the most widely accepted measure of endogenous insulin production (*32*). Relationships between abundances of classical and mono4 monocytes and stimulated C-peptide AUC were not detected among patients with ROT1D. However, we did identify a negative relationship between the baseline percentage of intermediate monocytes and the rate (slope) of C-peptide decline over the 24-month study period [(r (correlation coefficient) = −0.45, $P = 0.03$] (Fig. 8A). We also identified a positive relationship between the baseline percentage of nonclassical monocytes and the rate (slope) of C-peptide decline over the 24-month study period ($r = 0.46$, $P = 0.03$) (Fig. 8B). Over the study period, robust relationships between HbA1c or insulin dose and the abundances of classical, intermediate, nonclassical, or mono4 monocytes were not detected. However, positive relationships were measured between (i) the baseline HbA1c and baseline KLRD1^{high}/PRF1^{high} mono4 monocyte abundances as well as (ii) the 6-month insulin dose and baseline KLRD1^{high}/KLRB1^{high} mono4 monocyte abundances. A negative relationship was observed between the 24-month insulin dose and baseline nonclassical monocytes abundances (table S6).

Next, we examined the ratio between the percentage of intermediate monocytes versus the percentage of nonclassical monocytes. There was a large ratio range for the ROT1D group (Fig. 8C) that was, on average, higher than uHC ($P = 0.02$) and LST1D ($P = 0.02$). Notably among the ROT1D group, an inverse relationship was observed between the intermediate-to-nonclassical monocyte ratio at baseline and the percent change from baseline C-peptide AUC at 6, 12, and 24 months ($P < 0.02$; Fig. 8D). Similarly, the intermediate-to-nonclassical monocyte ratio at baseline was inversely related to the rate (slope) of C-peptide decline over the 24-month study period ($r = -0.62$, $P < 0.002$) (Fig. 8E). Together, these data support the hypothesis that (i) individuals with a higher intermediate-to-nonclassical monocyte ratio at onset will experience accelerated decline in β cell function; (ii) that intermediate monocytes promote more rapid post-onset progression; and (iii) nonclassical monocytes may be functioning in an immunoregulatory capacity to slow post-onset progression.

Longitudinal analysis of monocyte and T_{reg} populations in nonprogressors and T1D progressors

We previously reported that the circulating percentage of activated CD4$^+$CD45RA$^-$FoxP3^{high} T_{reg} increased in an age-dependent manner among HRS nonprogressors, suggesting an immunoregulatory mechanism underlying the decline in T1D susceptibility with age (*29*). However, among T1D progressors, the abundance of this highly suppressive subset remained stable. Using flow cytometry, we longitudinally examined PBMCs of nonprogressors and progressors for changes in circulating activated T_{reg} and monocyte abundances to better understand the temporal dynamics of these two populations.

Twenty-two HRS nonprogressors (mean age of 8.2 ± 3.5 years at the first sampling) were examined at two time points (mean duration of 5.4 ± 1.3 years between samplings). These stage 0 participants were either antibody negative at both time points ($n = 19$) or positive for a single antibody at one time point ($n = 3$). Between time points 1 and 2, there were changes in the abundances of classical (65.8 to 77.2%; $P = 0.001$), intermediate (20.9 to 13.8%; $P = 0.015$), and nonclassical monocytes (12.9 to 9.2%; $P = 0.061$) as well as changes in the intermediate: nonclassical ratio (3.5 to 2.0; $P = 0.045$) (Fig. 9A). Mono4 monocytes were defined as CD14$^+$ and/or CD16$^+$, HLA-DR$^+$, KLRD1^{high}, KLRB1^{high}, and PRF1^{high}. For these longitudinal analyses, the antibody panel was expanded to include intracellular staining for the cytolytic protein granzyme B (GZMB). Among classical, intermediate, and nonclassical monocytes of nonprogressors, significant temporal decreases in the percentage of KLRD1^{high}KLRB1^{high} (−41.5%, $P = 0.006$; −38.5%, $P = 0.009$; and −17.3%, $P = 0.031$, respectively; Fig. 9B), KLRD1^{high}PRF1^{high} (−15.8%, $P = 0.014$; −59.2%, $P = 0.049$; and −0.42%, $P = 0.019$, respectively; Fig. 9C), and KLRD1^{high}GZMB^{high} (−29.1%, $P = 0.045$; −32.8%, $P = 0.004$; and −13.5%, $P = 0.031$, respectively; Fig. 9D) cells were observed. When considering the total monocyte population (classical + intermediate + nonclassical), a significant temporal decline in the percentage of KLRD1^{high}KLRB1^{high} (−46.3%, $P = 0.001$), KLRD1^{high}PRF1^{high} (−37.8%, $P = 0.001$), and KLRD1^{high}GZMB^{high} (−54.1%, $P = 0.0001$) monocytes was also measured (Fig. 9, B to D). This contrasted with temporal increases in the ratio of peripheral activated CD4$^+$CD45RA$^-$FoxP3^{high} T_{reg} among total T_{reg} in these HRS (61.7%, $P = 0.0001$; Fig. 9E). Further, there was a significant temporal decrease in the ratio of mono4 monocytes (KLRD1^{high}KLRB1^{high}, KLRD1^{high}PRF1^{high}, and KLRD1^{high}GZMB^{high}) relative to activated T_{reg} among progressors (Fig. 9E). A significant negative relationship between activated T_{regs} (among total T_{regs}) and mono4 (among total monocytes) for these participants was observed when using any of these marker combinations (KLRD1^{high}KLRB1^{high}: $r = -0.69$, $P = 0.002$; KLRD1^{high}PRF1^{high}: $r = -0.77$, $P = 0.003$; and KLRD1^{high}GZMB^{high}: $r = -0.49$, $P = 0.002$).

We also conducted longitudinal analyses on 17 participants that advanced from stage 0 to stage 1 T1D or stage 2 T1D (as evidenced by presence of ≥2 islet autoantibodies, $n = 15$) or from stage 0 to stage 3 T1D (clinical onset, $n = 2$). The mean age of these participants was 9.0 ± 4.3 years, with a mean duration of 4.2 ± 2.3 years between samplings. In these progressors, we did not measure significant temporal changes in classical, intermediate, nonclassical, and mono4 monocyte abundances; or the ratio between the percentages of intermediate and nonclassical monocytes; or activated T_{reg} abundances (fig. S3, A to E). Overall, in contrast to T1D progression, these results indicate that nonprogression is associated with temporal decreases in mono4 monocytes, increased percentages of activated T_{reg} in the periphery, and decreases in the intermediate-to-nonclassical monocyte ratio.

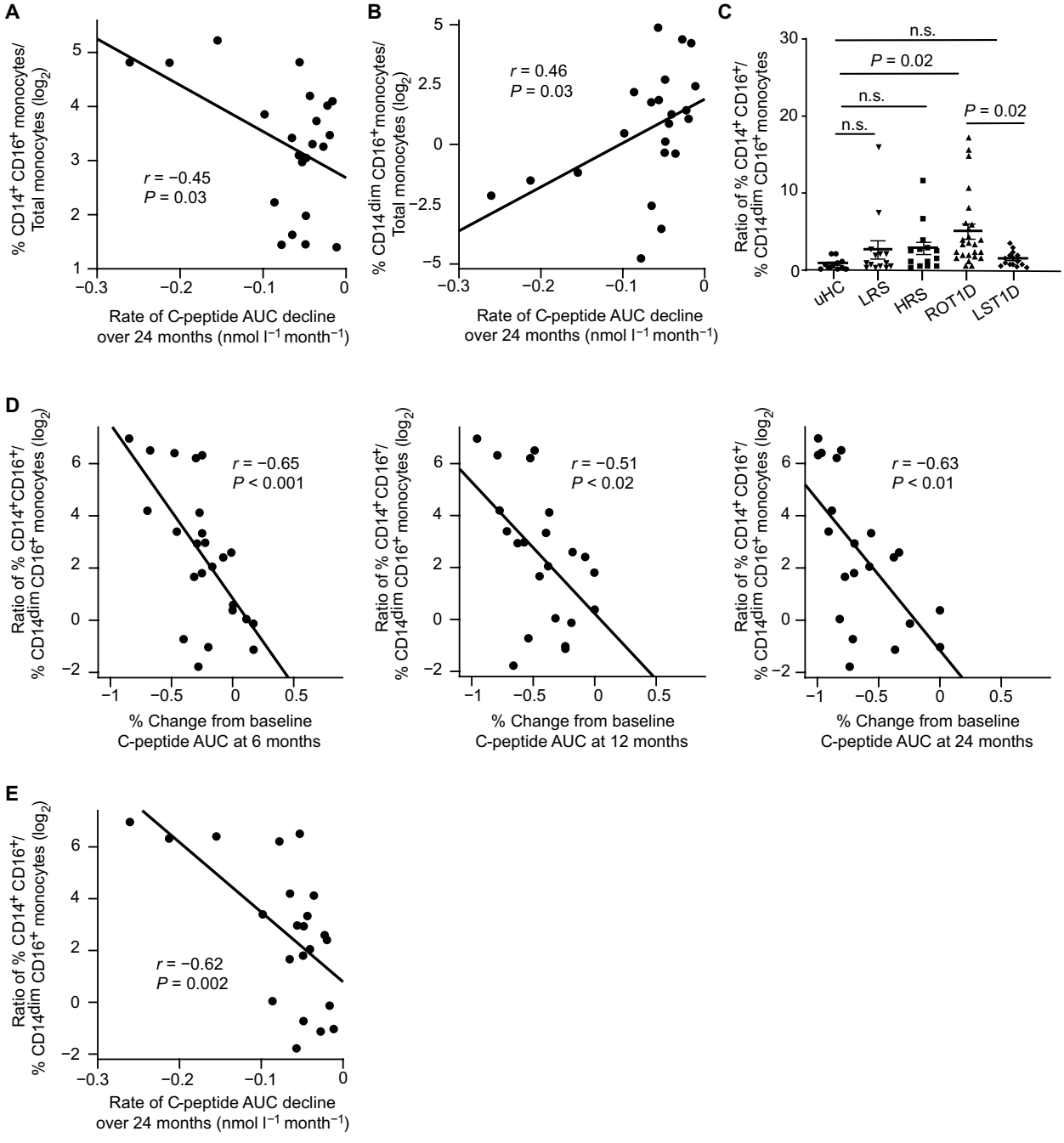


Fig. 8. The relationship between the percentages of intermediate and nonclassical monocytes at baseline and future β cell function among patients with ROT1D. (**A**) Significant inverse relationship exists between baseline percentages of $CD14^{+}CD16^{+}$ (intermediate) monocytes and the rate (slope) of C-peptide decline over the 24-month study period ($r = -0.45$, $P = 0.03$). The cohort consisted of 22 patients with ROT1D (age, 11.5 ± 3.3 years; HbA1c, 7.9 ± 1.5% at baseline). (**B**) Significant direct relationship exists between baseline percentages of $CD14^{dim}$ $CD16^{+}$ (nonclassical) monocytes and the rate (slope) of C-peptide decline over the 24-month study period ($r = 0.46$, $P = 0.03$). A truncated linear regression was used to estimate the slope of decrease for each individual. Larger negative slope values imply a faster decline of stimulated C-peptide. (**C**) Ratio of intermediate monocytes to nonclassical monocytes among the uHC, LRS, HRS, ROT1D, and LST1D cohorts illustrated in Fig. 6. (**D**) Significant inverse relationship exists between the intermediate-to-nonclassical monocyte ratio at baseline and 2-hour C-peptide AUC percent change from baseline at 6, 12, and 24 months after onset among patients with ROT1D. (**E**) Significant negative relationship exists between the intermediate-to-nonclassical monocyte ratio at baseline and the rate (slope) of C-peptide decline over the 24-month study period ($r = -0.62$, $P = 0.002$).

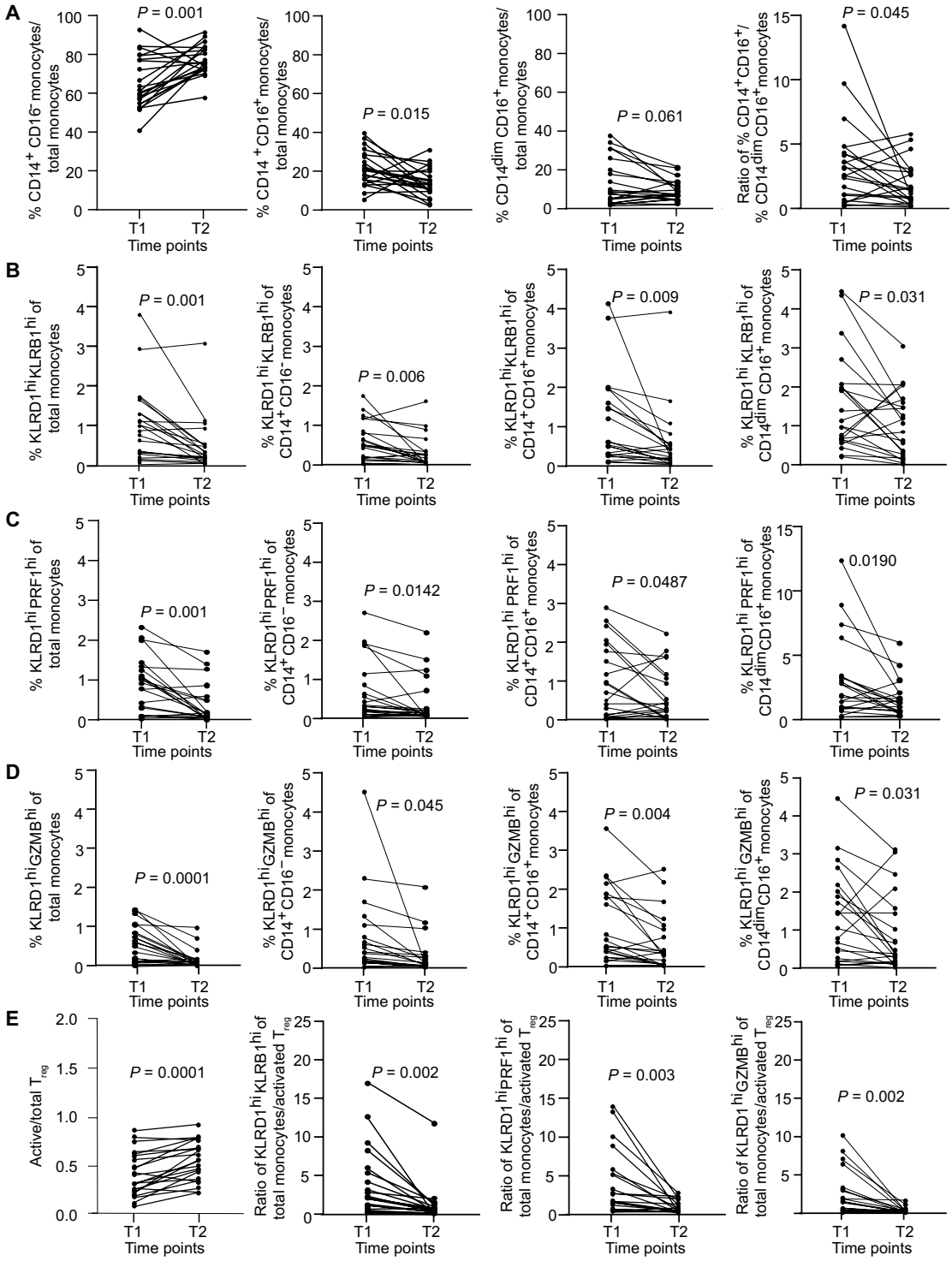


Fig. 9. Temporal analysis of monocyte and T_{reg} populations in stage 0 HRS nonprogressors. Twenty-two stage 0 HRS were analyzed at two time points. The mean age at the first sampling was 8.2 ± 3.5 years; the second sampling was conducted 5.4 ± 1.3 years later. (**A**) Percentages of KLRD1^{high}KLRB1^{high} monocytes among total, classical, intermediate, and nonclassical monocytes (left to right) at time points 1 and 2. (**B**) Percentages of KLRD1^{high}PRF1^{high} monocytes among total, classical, intermediate, and nonclassical monocytes (left to right) at time points 1 and 2. (**C**) Percentages of KLRD1^{high}GZMB^{high} monocytes among total, classical, intermediate, and nonclassical monocytes (left to right) at time points 1 and 2. (**D**) Representative flow cytometry profiles showing the gating strategy for resting and activated T_{reg} cells. Resting and activated CD4+ T_{regs} were defined as CD45RA^{+}FoxP3^{low} and CD45RA^{-}FoxP3^{high}, respectively. (**E**) Left to right: Abundance of activated T_{reg} among total T_{reg} at time points 1 and 2, ratio of KLRD1^{high}KLRB1^{high} monocytes among total monocytes: percentage activated T_{reg} among total T_{reg}, ratio of KLRD1^{high}PRF1^{high} monocytes among total monocytes: percentage activated T_{reg} among total T_{reg} at time points 1 and 2, and ratio of KLRD1^{high}GZMB^{high} monocytes among total monocytes:percentage activated T_{reg} among total T_{reg} at time points 1 and 2. Each dot represents a different individual. Horizontal bars indicate the mean ± SE. The differences between time points were assessed with a paired *t* test.

DISCUSSION

Monocytes comprise a substantial component of human peripheral blood, having essential functions in pathogen surveillance, phagocytosis, inflammation, and antigen presentation. While an understanding of monocyte heterogeneity and function in both health and disease has progressively increased, the number of studies that have examined monocytes in patients with T1D and their at-risk relatives remain limited.

Here, we determined that monocytes of T1D family members are hyperresponsive to TLR4 ligation, as evidenced by significantly higher IL-1β, TNF-α, and IL-6 secretion after exposure to LPS compared to uHC. This observation is consistent with prior reports describing hypersecretion of IL-1β and IL-6 by monocytes of patients with T1D regardless of whether they were stimulated with LPS or in an unperturbed resting state (*17*, *33*). These findings are relevant, as several clinical trials targeting proinflammatory cytokines have now provided proof of principle for these molecules as effectors in T1D by improving measures of β cell function. These trials include the targeting of TNF-α with etanercept (a TNF-α receptor fusion protein that fosters clearing of TNF-α from the circulation) and golimumab (an anti–TNF-α antibody) (*34*, *35*). Overall, trials of IL-1β blockade with anakinra (IL1-RA) or canakinumab (anti-IL1B antibody) did not slow β cell decline in ROT1D (*36*). However, subgroup analysis of the anakinra study detected significantly higher stimulated C-peptide levels among IL1-RA–treated participants having intermediate levels of β cell function at enrollment (*36*), and application of quantitative response analysis to the treatment arm of the canakinumab trial suggested a positive effect on retention of stimulated C-peptide (*37*). Further, our laboratory reported an inverse relationship between measured inflammation at onset and future stimulated C-peptide response in both anakinra- and canakinumab-treated patients when using a GO-based inflammatory index (*38*). A case for proinflammatory cytokines playing an effector role in T1D has grown through decades of study (*39*), and the hypersecretion of cytokines by monocytes in T1D families highlights the importance of innate immunity in T1D and its potential as a therapeutic target.

We also conducted gene expression studies of unstimulated monocytes that included scRNA-seq, where in addition to elevated LPS-stimulated cytokine expression, we associated this proinflammatory phenotype in T1D families with the expression of a cytolytic transcriptome that is consistent with expansion of the previously described mono4/NK-like subpopulation (*18*, *19*). Compared to uHC, cross-sectional flow cytometry analyses found that the mono4 population trended upward in healthy siblings of patients with T1D and was significantly elevated in patients with ROT1D. Longitudinal flow cytometry analyses of nonprogressors measured temporal decreases in the mono4 population that negatively correlated with increases in activated T_{reg}. This immunoregulatory pattern contrasted with T1D progressors, where neither temporal decrease in mono4 nor an age-dependent increase in active T_{reg} was observed.

Among patients with ROT1D, the abundances of classical, intermediate, nonclassical, or mono4 monocytes were not found to consistently correlate with HbA1c levels or insulin dose (table S6). Further, classical and mono4 monocyte abundances did not correlate with measures of stimulated C-peptide AUC. However, we determined that higher abundances of intermediate monocytes near clinical onset were associated with a more rapid decline in stimulated C-peptide over the subsequent 24-month study period (Fig. 8A). Ren *et al.* (*16*) reported a significant increase in nonclassical monocytes when comparing uHC to ROT1D (19.5 ± −9.1 days after onset). In contrast, we observed a significant decrease in the nonclassical monocyte percentage when comparing the ROT1D cohort (67.2 ± −34.2 days after onset) to the uHC cohort (Fig. 6B). This difference may stem from our sampling of participants later in the immediate post-onset period, when patients with ROT1D have generally achieved greater overall metabolic stabilization (*40*). However, our analyses of LST1D controls exhibiting similar levels of glycemic control suggest that the altered monocyte populations we observed in patients with ROT1D is not related to elevated glucose levels. We also identified a direct relationship between the baseline percentage of nonclassical monocytes and the rate (slope) of C-peptide decline over the 24-month study period (Fig. 8B). A more robust negative relationship was observed with rate (slope) of C-peptide decline when monocyte abundances were considered as a ratio between the percentages of intermediate versus nonclassical subsets. The ability to predict the persistence of C-peptide, particularly MMTT-stimulated values ≥ 0.2 nM, is relevant as patients that retain β cell function experience a reduced risk of severe hypoglycemia and microvascular complications (*41*–*45*). This observation may offer a biomarker of progression of immune-mediated β cell destruction and means to stratify patients. It further suggests an anti-inflammatory role for nonclassical monocytes in ROT1D, where their phagocytic removal of damaged cells and debris from the vasculature may function to promote wound healing and the resolution of inflammation (*9*). Similar to Hamers *et al.* (*46*), our scRNA-seq analysis identified a single $CD16^+$ subset; however, their mass cytometry studies revealed multiple nonclassical monocyte subsets that exhibited changes in frequency in the setting of severe coronary artery disease (*46*). Specifically, these authors identified a subset of $Slan^+$ nonclassical monocytes that they postulated to be anti-inflammatory and involved in the clearance of apoptotic cells (efferocytosis) and maintenance of vascular homeostasis (*46*). Whether subset differences among nonclassical monocytes exist in the context of T1D or T1D susceptibility is a limitation of this analysis that remains to be explored.

Similar to other studies that have used scRNA-seq to study monocyte populations, we observed heterogeneity extending beyond the traditional tripartite classification scheme. Villani *et al.* (*18*) sequenced 372 monocytes of two healthy participants to define four monocyte clusters, and Hu *et al.* (*19*) sequenced 11,259 monocytes of three healthy participants to define six clusters. In this study, we sequenced 13,915 monocytes of two uHC, two LRS, two, HRS and two patients with ROT1D to define 10 monocyte clusters. We were able to align 5 of 10 clusters with clusters identified by these prior studies (cluster 3 aligned with classical monocytes, cluster 5 aligned with nonclassical monocytes, cluster 0 aligned with S100A12 monocytes, cluster 4 aligned with HLA monocytes, and cluster 9 aligned with mono4/NK-like monocytes). The presence of additional unaligned clusters identified in this study may have stemmed from (i) the sequencing of a greater number of monocytes and (ii) analyzing primarily participants with T1D susceptibility (patients with T1D and unaffected siblings of patients with T1D). However, only the mono4/NK-like signature expressed by cluster 9, which was evident in the microarray studies, showed enrichment among monocytes drawn from the sibling cohorts with inherited T1D risk and even greater enrichment in the ROT1D cohort. Notably, not all

scRNA-seq studies of human monocytes have identified a mono4/NK-like monocyte subpopulation (*47*, *48*); authors of these studies have suggested that this transcriptomic signature is derived from physical doublets with NK cells. Here, steps were taken during scRNA-seq to minimize the inclusion of doublets, and we validated the presence of the mono4/NK-like monocyte population through a flow cytometric analysis that excluded $CD56^+$ NK cells and retained $HLA\text{-}DR^+$, $CD14^+$, $CD16^+$, $KLRD1^{high}$, $KLRB1^{high}$, and $PRF1^{high}$ monocytes. Villani *et al.* (*18*) observed that many monocytes defined as intermediate by surface markers coclustered in scRNA-seq analysis with the classical or nonclassical subsets. In line with this observation, we found the mono4 population distributed over the classical, intermediate, and nonclassical populations when applying a conventional rectangular gating strategy (Figs. 6 to 9).

While our studies find elevated abundances of mono4 monocytes in patients with ROT1D and find that their abundances decrease among nonprogressors, it is possible that the differentiation of monocyte subpopulations is altered in patients with T1D, and the expression of mono4 markers is a surrogate of this altered differentiation versus a driver of disease. We also recognize that the lack of longitudinal transcriptional analyses is a limitation of this study. However, in an elegant study, Xhonneux *et al.* (*49*) longitudinally analyzed whole blood transcriptomes of >400 T1D progressors and controls. The approach entailed RNA extraction from unfractionated whole blood, gene expression measurements by microarray, and a deconvolution approach, where the expression levels of immune cell–specific transcripts were used to estimate cell-type frequencies. The authors identified a signature consistent with NK cell enrichment that was significantly associated with both the development of islet antibodies and progression to clinical T1D (*49*). Here, we did not detect differences in the abundances of $CD56^+$ NK cell populations in our cross-sectional flow cytometric analysis of the uHC, LRS, HRS, LST1D, and ROT1D cohorts, nor in our longitudinal analyses of T1D progressors and nonprogressors. The NK signature defined by Xhonneux *et al.* (*49*) included 87 transcripts, notably 56 (64.4%; χ^2, $P < 1.0 \times 10^{-5}$) of these transcripts were included in the 826 transcript signature defined by our cross-sectional microarray analysis of negatively selected monocytes, and 30 (34.5%; χ^2, $P < 1.0 \times 10^{-5}$) of these transcripts were included in the top 100 transcripts of cluster 9. We analyzed the scRNA-seq data for differential expression of this 87 transcript signatures between uHC, LRS, HRS, and ROT1D participants for cells belonging to the NK cluster (fig. S4). Our findings support the conclusion that the mono4 (cluster 9) population defined here, as well as the expression of mono4 markers by monocyte populations defined by clusters 0 to 6 and 8 in patients with ROT1D (Fig. 4D), contributed to this prior observation.

Overall, we find that circulating monocytes in T1D families exhibit transcriptomes having elevated expression of cytolytic and proinflammatory mediators, hyper-responsiveness to LPS stimulation, and altered monocyte subset frequencies during progression and nonprogression. We observed increased mono4 abundances among patients with ROT1D and associated higher intermediate-to-nonclassical monocyte ratios with a more rapid decline in β cell functional mass during the post-onset period. We also found that nonprogression is associated with temporal decreases in mono4 monocytes over 5 to 6 years coincident with increases in the percentages of activated T_{regs} in the periphery and decreases in the intermediate-to-nonclassical monocyte ratio. These findings align with prior observations that associate T1D susceptibility and progression with a heightened immune response, a propensity for T_H1 responses toward insulin, and inflammation before the onset of autoimmunity [reviewed in (*50*)]. Further studies of monocyte subsets in the context of T1D progression/nonprogression may lead to new therapeutic strategies and improved biomarkers of pre- or post-onset disease progression.

MATERIALS AND METHODS

Experimental design

The objective of this study was to improve the understanding of monocyte function and heterogeneity among subjects with and at risk of developing T1D compared to uHC. Cross-sectional experimental assessments included measurement of cytokine release after LPS stimulation, transcriptomic studies, and confirmation by flow cytometry. The abundances of monocyte populations were correlated with measures of functional β cell mass in newly diagnosed patients and examined among at risk progressors and nonprogressors.

Study participants

Participants were recruited through Children's Wisconsin (CW) and diagnosis of T1D was defined per World Health Organization criteria (*51*). ROT1D samples were collected from participants collected 2 to 7 months after diagnosis from participants with histories of good glycemic control and had ≥1 autoantibody. Healthy control participants were free of known infection at sample collection and belonged to one of the following groups: (i) stage 0 siblings of probands with high-risk (DR3 and/or DR4) HLA genotypes (HRS), (ii) stage 0 siblings of probands with lower risk (non-DR3/DR4) HLA genotypes (LRS); and (iii) uHCs having no family history of T1D. The sibling controls were unrelated to each other and the patients with ROT1D studied. The LST1D cohort consisted of established patients with T1D where samples were collected 6.1 to 29.7 years after diagnosis. The study participants are described in table S1. The study procedures were approved by the CW Institutional Review Board (#01–15 and #1171017-6), and written informed consent was obtained from participants or their parents/legal guardians. Peripheral blood samples were drawn into acid citrate dextrose solution A or K+EDTA anticoagulant. Ficoll-Histopaque (Sigma-Aldrich, St. Louis, MO, USA) density gradient centrifugation was used to separate blood components immediately after draw, and plasma was stored at −80°C until use. Autoantibody titers (GAD, IA2, insulin, and zinc transporter 8) and class II HLA genotypes were determined as described (*52*, *53*). MMTTs were conducted within in 100 days of diagnosis (baseline) and then at 6, 12, and 24 months after onset to determine the rate of decline in stimulated C-peptide among patients with ROT1D as described (*32*, *54*).

LPS-induced secretion of cytokines

Monocytes were negatively selected from cryopreserved PBMCs using the EasySep Human Monocyte Enrichment Kit without CD16 depletion (STEMCELL Technologies Inc., Cambridge, MA, USA). The recovered population was devoid of NK cells, as measured by the absence of $CD56^+$ cells by flow cytometry. A total of 100,000 monocytes per well were cultured for 24 hours at 37°C with LPS (10 ng/ml) as described (*55*). The conditioned medium was recovered after centrifugation and stored at −80°C until IL-1β, IL-6, and

TNF-α levels were measured by enzyme-linked immunosorbent assay (R&D Systems, Minneapolis, MN, USA).

Affymetrix GeneChip analyses of negatively selected monocytes

RNA was extracted from monocytes negatively selected from PBMCs as described above. Transcriptomes were measured using Affymetrix GeneChip Human Genome U133 plus 2.0 arrays (Affymetrix, Santa Clara, CA, USA) as described (*29*). Array data were subjected to global median normalization with Bioconductor Robust Multi-array Analysis (*56*). The significance of gene expression measurements was determined by analysis of variance (ANOVA), and the rate of type I errors in multiple testing was assessed by calculating FDR with Partek Genomics Suite 7.2 (Partek, Saint Louis, MO, USA). Ontological analyses were conducted with the Ingenuity Pathway Analysis package (Ingenuity Systems, Redwood City, CA, USA) and the Database for Annotation, Visualization, and Integrated Discovery version 6.7 (*57*).

scRNA-seq of sorted PBMCs

Cryopreserved PBMC samples of eight age-matched participants (two well-controlled patients with ROT1D, two HRS, two LRS, and two uHC) were thawed and incubated with the fixable LIVE/DEAD Violet dye (Life Technologies, Grand Island, NY, USA) for 30 min on ice. The cells were subsequently stained with antibodies targeting CD14 (clone M5E2, BD Biosciences, Franklin Lakes, NJ, USA), CD16 (clone 3G8, BD Biosciences), HLA-DR (clone G46-6, BD Biosciences) for 30 min on ice and then washed and passed through a 30 μM filter. PBMCs were sorted with a FACSAria III cell sorter (BD Biosciences), with specific forward and side scatter settings to exclude doublets. Using a hemocytometer and trypan blue staining, the recovered HLA-DR^{high}, CD14^{+}, CD16^{+}, and HLA-DR^{low}, CD14^{-}, and CD16^{-} cells were determined to have >40,000 cells per sample and viability exceeding 90%. These two populations were mixed in a 1:1 ratio to generate a monocyte-enriched, single-cell, PBMC population. Single-cell Gel Beads-in-emulsions were prepared with the Chromium Single Cell 3'v3 Reagent Kit (10X Genomics, Pleasanton, CA) and a Chromium instrument. Single-cell capture, barcoded reverse transcription, cell lysis, and library preparation were performed as per the manufacturer's protocol. Library sequencing used a NovaSeq 6000 S2 Reagent kit (100 cycles; Illumina, San Diego, CA, USA) and a NovaSeq Sequencing System (Illumina).

scRNA-seq data analysis

Raw sequencing data were processed using the 10X Genomics Cell Ranger software (version 3.0.0) for sample demultiplexing, alignment to the human GRCh38.p14 reference, generating cell barcode, and gene unique molecular index (UMI) count matrices. Preprocessing of the scRNA-seq data and downstream analysis was performed using the Seurat package (version 4.3.0) (*58*) in R. To filter low-quality cells or potential doublets, cells expressing fewer than 200 or more than 5000 genes or cells with more than 20% of UMI counts coming from mitochondrial genes were removed from the analysis. Centered log ratio transformation was used as the normalization method, and the 99% quantile was set as the threshold to declare if a cell was considered positive. After quality control, a total of 38,594 cells were retained for downstream analysis.

To integrate samples belonging to the four cohorts, we first log-normalized the count data and identified the top 2000 most variable transcripts (using method vst) for each cohort. We used a canonical correlation analysis–based algorithm to remove batch effect across four cohorts (by using function IntegrateData). Integrated data were further scaled with removal of variation from total UMI counts and percentage of mitochondrial genes. To reduce dimensionality and denoise the data, PCA was performed on the top 2000 variable genes. Shared nearest neighbor of each cell was constructed using first 30 PCs, and different cell clusters were identified using the Louvain-Jaccard graph-based algorithm. We used t-SNE with 30 PCs as input to visualize the data in two-dimensional space along with cluster results from the clustering algorithm. Identification of differentially expressed genes between groups of cells was performed by Wilcoxon rank sum tests. Cluster marker genes were declared with adjusted *P* value less than 0.05 (with Bonferroni correction for multiple testing) and were used to define well-known PBMC populations.

A recluster analysis focusing on monocytes was performed by first retrieving the cells from clusters 0 and 7 (Fig. 3), the aforementioned normalization was reperformed, and the batch correction method was used to integrate data from the four cohorts. This was followed by dimension reduction through PCA using the first 10 PCs as input for UMAP to visualize the data in two-dimensional space. Annotations were confirmed with R package Azimuth (*25*) using the human PBMC reference. Pseudotime trajectory analysis was performed with R package Monocle3 (*59*).

Dimension reduction plots (t-SNE and UMAP), heatmaps, violin plots, dot plots, and feature plots were generated using functions DimPlot, DoHeatmap, VlnPlot, DotPlot, and FeaturePlot in Seurat, respectively. Bar plots were generated using the R package ggplot2 (version 3.4.2). The AddModuleScore function in Seurat was used to calculate gene module scores.

Analysis of monocyte subsets by flow cytometry

Cryopreserved PBMCs (~3 × 10^6 PBMCs per participant) were thawed, then incubated with the fixable LIVE/DEAD Violet dye (Life Technologies, Grand Island, NY, USA) for 30 min on ice, and then stained with fluorochrome-conjugated monoclonal antibodies targeting CD3 (clone UCHT1, BD Biosciences), CD19 (clone HIB19, BD Biosciences), CD14 (clone M5E2, BD Biosciences), CD16 (clone 3G8; BD Biosciences), HLA-DR (clone G46-6, BD Biosciences), CD56 (clone B159, BD Biosciences), KLRB1/CD161(clone HP-3G10, BD Biosciences), and KLRD1/CD94 (clone HP-3D9, BD Biosciences) on ice for 30 min. Next, cells were washed, fixed, and permeabilized for intracellular staining using the FIX & PERM Cell Permeabilization Kit (GAS004, Thermo Fisher Scientific) in accordance with the manufacturer's protocol. Then, PBMCs were stained with anti-PRF1 (clone B-D48, BioLegend), and anti-GZMB (clone GB11, BD Biosciences) for 45 min. T_{reg} populations were stained as previously described (*29*) using anti-CD4 (clone RPA-T4), anti-CD25 (clone M-A251), anti-CD45RO (clone UCHL1), anti-CD45RA (clone HI100), anti-CD127 (clone HIR-7R-M21) (BD Bioscience, San Jose, CA), and anti-FOXP3 (clone PCH101) (eBioscience, San Diego, CA). Representative flow cytometry gating plots for the T cell staining protocol are shown in fig. S5.

The stained PBMCs were analyzed on a Cytek Aurora flow cytometer (Cytek Biosciences) using five laser line configurations, blue (488 nm), red (640 nm), yellow green (561 nm), violet (405 nm) and ultraviolet (355 nm) with Spectro Flo software v2.2.0.2, which automatically calculated compensation by unmixing the cells and correcting spectral spillover using the single stained reference controls

for each fluorescent tag, and unstained control with autofluorescence extraction option was enabled. Further, to minimize the unmixing error, the spectrum profiles for all single stained reference controls were reconciled against the expected spectrum profiles provided by the Cytek Full Spectrum Viewer (https://spectrum.cytekbio.com/).

Supplementary Materials

This PDF file includes:
Figs. S1 to S5
Legends for tables S1 to S6

Other Supplementary Material for this manuscript includes the following:
Tables S1 to S6

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